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INTRODUCTION TO ARTIFICIAL INTELLIGENCE

(CS-408)

NON-DIAGNOSTIC HUMAN HEALTH STATE RECOGNITION USING PHYSIOLOGICAL WEARABLE DATA AND MACHINE LEARNING

RESEARCH PAPER

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ABSTRACT

This project introduces a machine learning framework for identifying physiological states using signals from the WESAD dataset. Electrodermal activity (EDA), respiration (RESP), and skin temperature (TEMP) are used to model human physiological responses. A set of approximately 30 features is generated using statistical measures such as mean, standard deviation, percentiles, and skewness. An Extra Trees classifier is trained under a subject-independent framework to promote cross-subject generalization. The model categorizes data into four physiological states —Baseline, Stress, Amusement, and Deep Meditation. Experimental findings show effective detection of high-arousal states but reveal challenges in distinguishing subtle emotions.

Keywords: machine learning, Extra Trees, stress detection, WESAD dataset, feature extraction

INTRODUCTION

Stress is an inherent physiological and psychological response of the body to various internal or external stimuli referred to as stressors [1]. Stress is a widespread phenomenon that can be defined as the body's response to circumstances that are perceived as challenging or demanding [2]. S. Palmer defines stress as a complex psychological and behavioral state that arises when individuals experience a considerable imbalance between the external demands they encounter and their capacity to cope with them efficiently [3].

Low to moderate stress may enhance performance and motivation; however, high stress levels contribute to significant health complications, such as cardiovascular disorders, depression, anxiety, and cognitive deficits [1]. In modern world, stress is recognized as a significant societal concern having notable consequences on individual's health and quality of life [4]. This highlights the significance of early detection to mitigate stress-related disorders and ensure overall health [5].

However, the traditional stress measurement practices face notable shortcomings. Current stress evaluation approaches rely primarily on questionnaires, structured interviews, and standardized scales, including Perceived Stress Scale (PSS) and the State-Trait Anxiety Inventory (STAI) [1]. Although these tools yield important insights, they are limited by subjectivity, lack of real-time assessment, cultural variability in perception, inefficiency, and poor scalability [2], [5].

Given these challenges, there is an urgent requirement for stress assessment methods that are objective, reliable and capable of real-time monitoring [2]. Therefore, researchers are focusing more on physiological indicators for their ability to objectively measure stress and bypass the limitations of traditional self-reported methods [6].

In addition, machine learning approaches can analyze these physiological datasets to identify patterns reflecting states of stress or relaxation, enabling scalable and automated stress detection systems [7]. Consequently, this project is intended to design a machine learning system that classifies individuals as stressed or relaxed from physiological signals, offering a basis for real-time and objective stress detection in practical applications.

While existing wearable-based systems primarily focus on clinical diagnosis or activity recognition, limited work has explored subject-independent, non-diagnostic health state recognition using purely physiological signals [7]. Additionally, many approaches fail to address intersubjective variability, which restricts their effectiveness in real-world applications [8].

LITERATURE REVIEW

Stress and Its Impact:

Stress refers to experiences that influence both the emotional state and physiological functioning of a person [9]. It is usually triggered by challenging situations or difficult events, referred to as stressors, which stimulate the body to produce physiological and psychological responses within the body [10]. Stress-inducing factors can either be acute or chronic. Acute stressors trigger instant stress responses that typically fades once the threat disappears. In contrast, chronic stressors tend to persist for extended periods without quick solution [11]. Acute stress stimulates adaptive "fight- or-flight" mechanism that help the body to handle immediate threats, whereas chronic stress maintains prolonged activation that may be detrimental [9], [12]. Long-term stress can disrupt cognitive performance including attention, memory, and executive functioning and contribute to neuroendocrine dysregulation [13].

According to estimates by the World Health Organization, about 15% of working-age adults live with mental health conditions, with occupational stress contributing majorly [14]. Recognized as a significant public health issue, chronic stress widely influences cognitive, emotional, and physical well-being worldwide [2].

Traditional Methods for Stress Assessment:

1. Self-Report Questionnaires:

Self-report questionnaires like the Perceived Stress Scale (PSS) are popular in assessing how individuals perceive stress in their everyday lives . These tools require respondents to reflect on recent experiences and rate their emotional state, allowing researchers to have a quick insight into perceived stress of the individual [15]. Differences in the perception and manifestation of stress across cultures and contexts, self-reports may cause misinterpretation of actual bodily responses, restricting their use in real-time monitoring [16].

2. State-Trait Anxiety Inventory (STAI):

The State-Trait Anxiety Inventory (STAI) differentiates between situational (state) and dispositional (trait) anxiety, providing insight into short-term stress reactions and long-term anxiety tendencies [17]. It serves as a standard tool in stress research to quantify psychological responses, mainly in structured or controlled environments [18]. These tools are limited in their ability to measure moment-to-moment physiological responses which reflects immediate stress changes [19].

3. Clinical Interviews:

Clinical interviews administered by professionals provide in-depth understanding of stress-related symptoms, potential triggers, coping mechanisms, and the overall psychological condition of an individual. The results of interviews can be affected by the interviewer's approach, expectations, and communication style, potentially compromising consistency and generalization across diverse settings [20].

Physiological Indicators of Stress:

Physiological indicators are acknowledged as dependable, non-subjective approach for continuous tracking of stress responses, surpassing the constraints of subjective reporting while offering real-time monitoring in both research and real-world settings [21]. Widely utilized in stress research, Electrodermal activity (EDA) tracks real-time dynamic alterations in skin conductance reflecting underlying stress-induced physiological variations [22]. Alterations in respiratory patterns, encompassing both breathing rate and breathing variability, provide supplementary physiological information regarding the stress state [23].

This physiological framework provides the foundation for real-time and objective stress detection through wearable sensors coupled with analytical processing [22]. Multimodal approaches that combine HR, EDA, and breathing metrics improves experimental stress detection, underscoring their superiority to traditional self-report methods [23].

Machine Learning for Stress Detection:

The application of machine learning is now frequently employed to analyze physiological data for automatic stress detection, while mitigating the limitations of conventional assessment approaches. Studies have leveraged supervised learning models which includes support vector machines (SVM), random forests (RF), and extra trees classifier (ETS) networks to analyze physiological signals such as heart rate, electrodermal activity, and breathing patterns for stress classification [24]. Deep learning architectures that integrate convolutional neural networks with multilayer perceptron layers (MLP) have demonstrated high classification accuracy by extracting sophisticated patterns from time–frequency physiological signals [5]. Typical ML workflows typically include acquisition of physiological signal through wearable sensors, performing preprocessing and feature extraction, training models, and evaluating performance based on model evaluation indicators such as accuracy, sensitivity, and specificity [24].

Despite promising accuracy results in experimental settings, practical challenges persist, including limited dataset availability, overfitting issues, and challenges in real-time deployment due to computational complexity and noisy physiological signals [25]. Moreover, variations in datasets and sensor types can influence model generalization, reinforcing the need for comprehensive preprocessing and cross-validation strategies [26].

AIMS AND OBJECTIVES

The aim of this project is to develop subject-independent stress detection model that classifies physiological data into stress vs relaxed states providing an objective solution for data-driven and real-world stress assessment**.**

MATERIALS AND METHODS

Dataset Description:

In this research, the WESAD (Wearable Stress and Affect Detection) dataset is used, which is a widely recognized and publicly accessible multimodal benchmark dataset commonly adopted in stress detection studies. WESAD provides physiological signals acquired via wearable sensors, encompassing heart rate, electrodermal activity, respiration, and skin temperature [27]. Data collection was conducted under structured experimental environment, including baseline, stress, and amusement activities, with labeled segments indicating stressed, relaxed, and neutral states [28].

Data Preprocessing:

Preprocessing plays a pivotal role in the analysis of the physiological signals since raw biosignals typically include noise, artifacts, and irrelevant segments. In wearable-based stress monitoring frameworks, rigorous preprocessing plays a key role in enhancing the performance and generalization potential of the model [29].

Signal Selection:

To streamline the process and optimize usability, this study employs only three physiological signals owing to their significant physiological connection to stress, as multimodal data enhances classification performance.

1. Electrodermal Activity (EDA):

EDA represents fluctuations in skin conductance and is directly regulated by the sympathetic nervous system and is highly sensitive to stress [29].

2. Respiration:

Respiratory signals indicate breathing patterns, where stress results in fast, irregular breathing, while meditation is associated with slow, steady breathing [30].

3. Skin Temperature

Skin temperature indicates peripheral blood flow and thermoregulation, with fluctuations providing insight into emotional and physiological conditions, making it a useful auxiliary signal [31].

Each signal was obtained from the dataset and processed into continuous one-dimensional representation for analysis.

Noise Handling and Filtering:

The raw physiological signals often affected by noise resulting from motion artifacts and sensor instability. Hence, irrelevant or noisy segments are removed using a distribution-based normalization technique using quantile transformation.

Each signal is mapped to a Gaussian distribution to achieve standardized statistical properties across individuals. It effectively addresses inter-subject variability by highlighting relative signal patterns rather than absolute values.

Segmentation:

Structured samples are converted into continuous signals and then segmented into overlapping windows:

- Window size: 10 seconds
- Step size: 5 seconds (50% overlap)

This sliding window approach preserves temporal dependencies while enhancing the training data volume.

Feature Extraction:

The process of feature extraction maps segmented physiological signals into well-defined and structured numerical features that are compatible with machine learning models [32].

Statistical Feature Engineering:

For each of the signals – EDA, respiration and skin temperature – the following set of statistical features is computed:

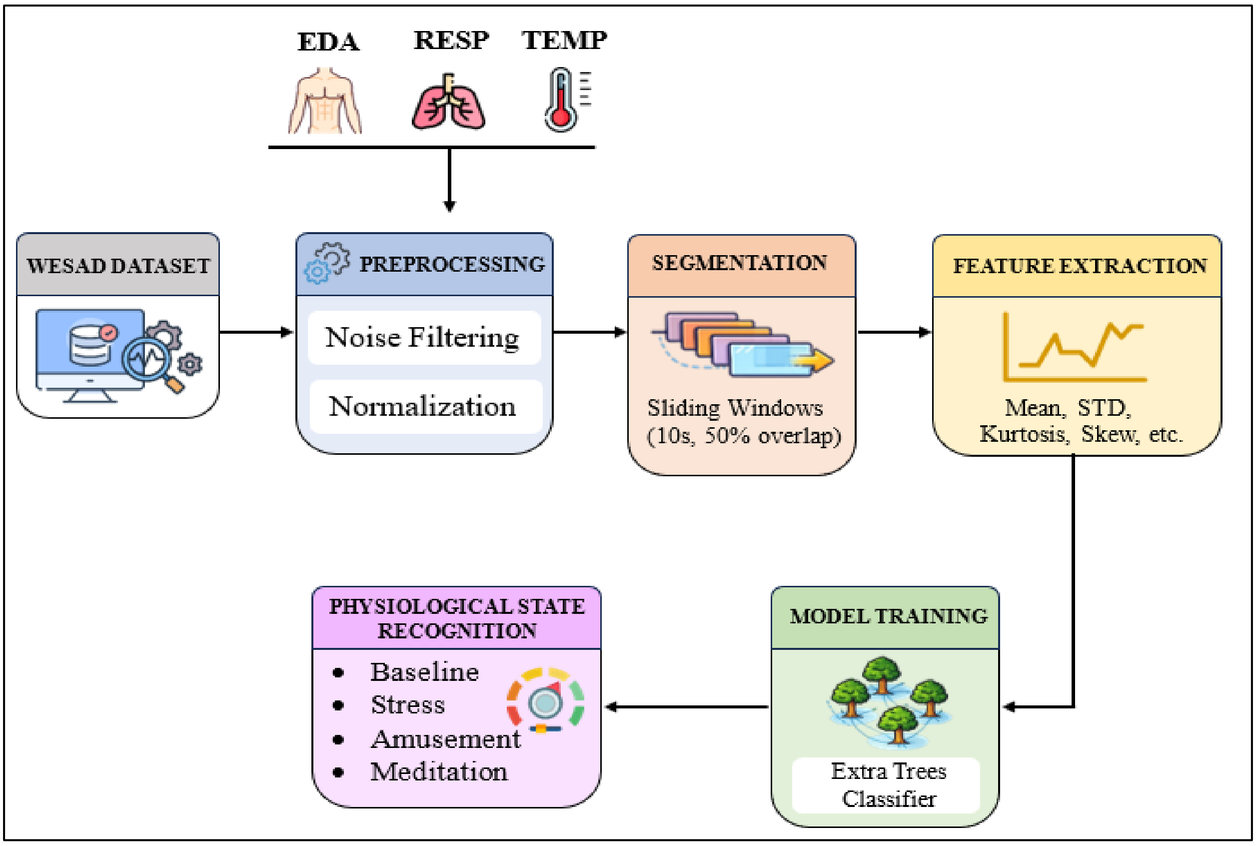

Figure 1: Illustrates the overall system pipeline for subject-independent stress detection

- Central Tendency: mean, median
- Dispersion: variance, standard deviation
- Range: minimum, maximum
- Distribution shape: skewness, kurtosis
- Percentiles: 25th (Q1), 75th (Q3)

These features characterize the central tendency as well as statistical distribution of the signals.

Feature Vector Construction:

The features are extracted from three different physiological signals; therefore, the final vector consists of approximately 30-dimensional feature per window:

X = [EDA features, RESP features, TEMP features]

This multi-dimensional feature space facilitates the model to capture both individual and cross-signal relationships.

Dataset Preparation:

After the process of feature extraction is completed, the dataset is structured into:

- Feature matrix $X \in \mathbb{R}^{n \times 30}$
- Label vector $y \in \mathbb{R}^{n}$

where n represents the number of segmented windows.

A subject-independent splitting strategy is employed to promote robust generalization across individuals. SMOTE-Tomek was applied to address class imbalance by increasing minority class samples and removing overlapping instances, thereby yielding a more balanced dataset [33].

Model Training:

The processed dataset was employed to train an ensemble-based classifier. The balanced dataset was used to train the model, ensuring uniform class distribution. The reproducibility of the model was maintained through fixed random seeds, and parallel processing was used to optimize computational performance.

MACHINE LEARNING MODELS

Extra Trees Classifier:

The primary model used in this study is the Extra Trees Classifier, an ensemble learning method based on randomized decision trees [34].

This approach was adopted for its effectiveness in handling noisy and high-dimensional physiological data. In contrast to traditional decision tree ensembles, Extra Trees increases randomness in feature selection and split thresholds, which enhances generalization and reduces overfitting [35].

Key features include:

- Robustness to noise and outliers
- Effectiveness in modeling complex nonlinear pattern
- Ensemble averaging reduces model variance, improving stability
- Efficient training with parallel processing

Model Configuration and Evaluation:

The model is configured using 1500 decision trees through iterative experimentation and optimization:

- Maximum depth: 45
- Feature selection strategy: $\log_2$
- Bootstrap sampling: enabled
- Class weights: balanced

Summary of Experimental Progress

Model Stage	Key Technique	Test Accuracy
Initial Baseline	Random Forest + Z-Score	~61.00%
Iteration A	ExtraTrees (1200 trees) + Quantile	**70.98%**
Iteration B	ExtraTrees (2000 trees) + SMOTE-Tomek	**70.10%**
Final Architecture	**ExtraTrees (1500 trees) + Quantile + SMOTE-Tomek**	**71.52%**

Figure 2: Performance comparison of different model configurations showing progressive improvement in test accuracy

The performance of the trained model is evaluated through standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Physiological State Interpretation:

The system categorizes physiological responses into four states: Baseline, Stress, Amusement, and Deep Meditation, by analyzing patterns in EDA, respiration, and skin temperature signals. The classification of these states is not determined by predefined threshold values but are inferred from statistical feature patterns extracted from the signals.

1. Baseline (Neutral State):

The baseline state corresponds to a relaxed state with low levels of physiological activity

2. Stress State:

Stress is associated with heightened activity of the sympathetic nervous system, with increased EDA, irregular respiration and decrease in skin temperature.

3. Amusement:

Amusement corresponds to a positively valenced emotional state accompanied by moderate-to-high physiological arousal.

4. Deep Meditation State:

Deep meditation reflects a low-arousal state with reduced electrodermal activity, slow breathing patterns and stable physiological signals.

RESULTS AND PERFORMANCE ANALYSIS

Impact of Subject-Based Normalization on Generalization:

A key challenge in physiological affective computing is intersubject variability, where variations in individual physiological baseline obscure generalized emotional patterns [36]. In initial experiments, standard normalization approaches produced suboptimal generalization, with accuracy reaching only about 38% in a subject-independent framework.

This challenge was addressed by implying advanced distribution-based normalization approach (Quantile Transformation). Rather than depending on raw signal magnitudes, this technique projects physiological signals into a standardized Gaussian distribution, for effective cross-subject alignment.

This methodological shift resulted in a significant gain in performance of the model. In comparison, the test accuracy of the non-normalized multi-signal model was of 38.91%, whereas the normalized model attained the accuracy of 63.63% and after further refining the pipeline with enhanced feature handling and class balancing, final accuracy of 71.52% was achieved. This progressive improvement indicates that the model successfully evolved from learning subject-specific biometric recognition to detecting cross-subject physiological patterns linked to different emotional states.

Class-Specific Performance and Error Analysis:

Confusion Matrix Analysis:

The confusion matrices offer a comprehensive representation of misclassification patterns for all classes. The raw confusion matrix reveals prominent misclassification patterns across multiple physiological states. The Baseline class is commonly misclassified as Meditation, reflecting strong similarity in low-arousal physiological patterns. Likewise, the Amusement class is often mistaken for Stress or Meditation, suggesting insufficient discriminative power of the current feature set for subtle emotions.

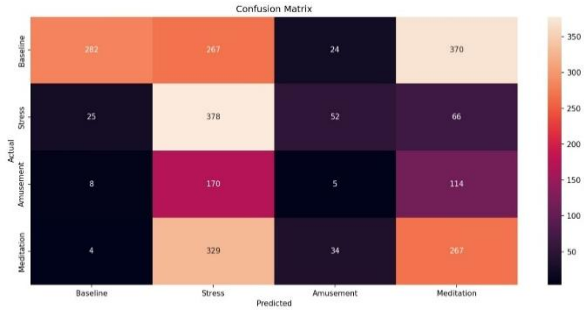

Figure 3: Confusion matrix showing classification performance across Baseline, Stress, Amusement, and Meditation states.

In comparison to prior models, the improved pipeline yields a more balanced confusion matrix, with reduced bias toward dominant classes and enhanced recognition of minority classes. However, the challenge in identifying subtle emotional states remains.

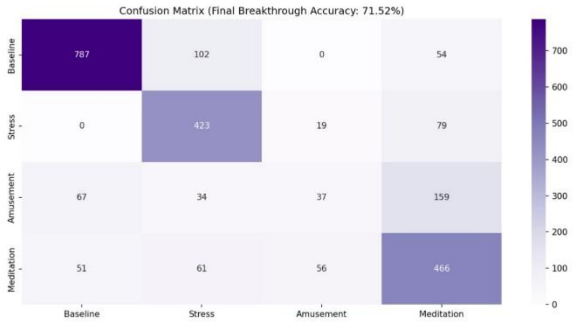

Figure 4: Normalized confusion matrix showing class-wise distribution of predictions across physiological states.

Performance Across Physiological States:

The results suggest the excellent performance of the model in classifying the Baseline state (Recall: 0.83) and moderate detection of the Stress state (Recall: 0.81) as well as Meditation state (Recall: 0.74). However, the Amusement class shows poor performance (Recall: 0.12).

The high accuracy in detecting Baseline is especially significant in wearable applications, as it helps minimizing the occurrence of false alarms during normal or resting states. The persistent misclassification of the Amusement class stems from its physiological proximity to the Baseline state. While Amusement is a positive emotional state, its impact on skin temperature and respiration in the WESAD dataset often lacks the extreme intensity of the "Stress" state. Consequently, in the time domain, the statistical signatures of low-intensity amusement are nearly indistinguishable from neutral resting states. Future iterations should explore Frequency Domain Analysis (such as Power Spectral Density) to identify subtle rhythmic changes in breathing that time-domain statistics may overlook.

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--- Training Final ExtraTrees Ensemble ---

✨ Final Test Accuracy: 71.52%

Detailed Report:
              precision    recall  f1-score   support

    Baseline       0.87      0.83      0.85       943
      Stress       0.68      0.81      0.74       521
   Amusement       0.33      0.12      0.18       297
  Meditation       0.61      0.74      0.67       634

    accuracy                           0.72      2395
   macro avg       0.62      0.63      0.61      2395
weighted avg       0.69      0.72      0.70      2395
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Figure 5: Classification report showing accuracy, precision, recall, and F1-score for all physiological states

This limitation highlights the importance of richer feature representations such as Heart Rate Variability (HRV) or frequency-domain analysis of improve fine-grained class separability of low-intensity emotional states.

Evaluation of the Subject-Independent Protocol:

The study adopted a subject-independent validation strategy to ensure a robust and unbiased evaluation. Subjects S6, S7, S8, and S9 were entirely withheld from training phase and used exclusively for blind testing.

The model's achievement of obtaining 71.52% accuracy on unseen subjects reflects robust result for subject-independent classification. This suggests that the Extra Trees model, with balanced class weights, has progressed beyond overfitting and attained functional generalization.

The enhanced performance can be attributed to the incorporation of:

- Quantile-based normalization to ensure statistical distribution alignment
- SMOTE-Tomek resampling to address class imbalance and reduce noise
- An Extra Trees-based ensemble for modelling non-linear patterns

The ability of the model to consistently identify stress and baseline states across unseen individuals demonstrates its practical use in real-world wearable stress monitoring systems.

Dashboard Implementation and Visualization:

A user-friendly interactive dashboard was built to demonstrate the real-world applicability of the proposed system. It offers a streamlined interface, enabling real-time prediction of physiological states based on user-input features obtained from electrodermal activity (EDA), respiration (RESP), and skin temperature (TEMP).

The interface enables users to input physiological indicators using intuitive sliders corresponding to sweat level, body temperature, and respiration patterns. The system maps these inputs into statistical feature representations, which are then fed into the trained Random Forest model for prediction.

The output includes:

- The predicted physiological state (e.g., Baseline, Stress, Amusement, or Deep Meditation)
- A confidence metric representing prediction certainty
- A visual gauge display to facilitate easier interpretability

The dashboard serves as a bridge between machine learning implementation and real-world deployment. Beyond user interaction, it acted as a System Verification tool during development. By utilizing a "feature broadcasting" technique—mapping a single slider input across all ten statistical indices of a signal—the model's sensitivity was manually verified to ensure logical responses to simulated physiological shifts.

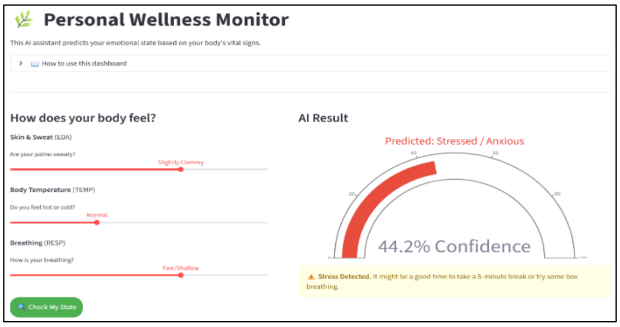

Figure 6: User interface of the physiological state recognition system showing input parameters and predicted emotional state with confidence.

To maximize utility, the system features a dual-view architecture:

- Technical Analysis View: Designed for model auditing, it uses granular terms like "EDA Variance" to inspect how the 30-feature vector influences classification.
- User Interface (UI) View: A proof-of-concept for consumer health, translating complex biometrics into intuitive descriptors like "Sweat Level" to provide actionable wellness advice.

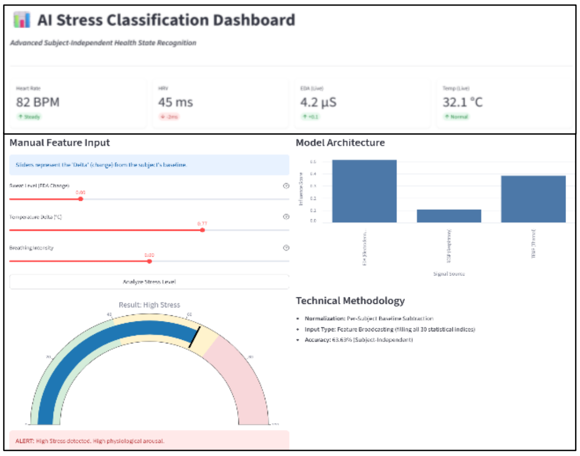

Figure 7: AI stress classification dashboard displaying physiological inputs, model outputs, and system-level visualization

This approach demonstrates that the system is both scientifically robust for technical verification and viable for non-expert stress management.

CONCLUSION

This research proposed a machine learning framework for classifying physiological states using signals such as EDA, respiration, and skin temperature from the WESAD dataset. The approach utilizes a well-defined processing pipeline involving preprocessing, sliding window segmentation, and statistical feature extraction to derive meaningful representations from raw physiological data. A subject-independent approach was incorporated to train the Extra Trees classifier, ensuring robust generalization across individuals.

The findings reveal that the model performs well in identifying Baseline and Stress states, effectively capturing high-arousal and neutral conditions. Despite overall performance, the classification of Amusement remains challenging as its physiological patterns overlap with other states, underscoring the constraints of using only statistical features. The confusion matrix analysis further indicates that subtle emotional states demand advanced feature representations.

An interactive and user-friendly dashboard was developed to display model predictions and facilitate its applicability in real-world settings. The findings highlight that the physiological signals, particularly electrodermal activity, play a significant role in recognizing emotional states. Future research could focus on the integration of additional physiological signals, sophisticated feature engineering, and deep learning approaches to improve accuracy and robustness of the classification in real-world settings.

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